

TAMIL NADU STATE BOARD - STANDARD 11 MATHEMATICS VOLUME 2

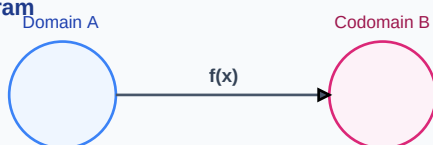
CHAPTER 12: INTRODUCTION TO PROBABILITY THEORY

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Mathematics Volume 2
Chapter Name:	Introduction to Probability Theory

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Conditional Probability	The probability of an event A occurring given that another event B has already occurred, denoted $P(A B)$.
Independent Events	Two events are independent if the occurrence of one does not affect the probability of occurrence of the other.
Sample Space	The set of all possible outcomes of a random probability experiment, denoted by S.

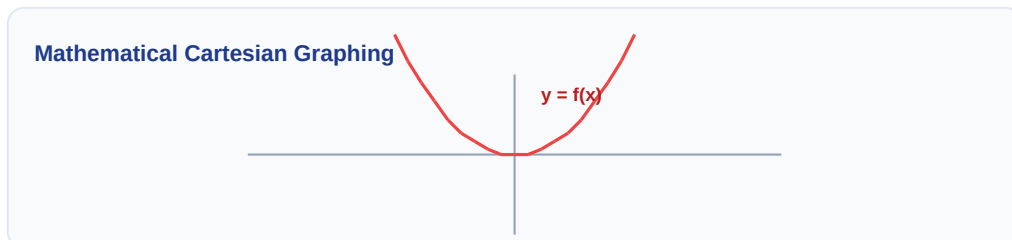
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Conditional Probability: $P(A | B) = P(A \cap B) / P(B)$
- Addition Theorem: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Bayes' Theorem: $P(A_i | B) = [P(B | A_i) * P(A_i)] / \sum [P(B | A_j) * P(A_j)]$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: If $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, find $P(A | B)$.

Step-by-step Solution:

Apply conditional probability: $P(A | B) = P(A \cap B) / P(B) = 0.3 / 0.5 = 3/5 = 0.6$.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: A bag contains 5 red and 3 blue balls. If two balls are drawn at random without replacement, find the probability that one is red and one is blue.

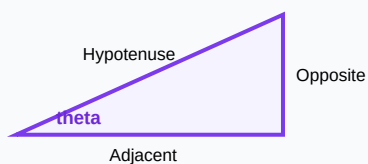
Analytical Solution Strategy:

Total balls = 8. (1) Red first, then Blue: $(5/8) * (3/7) = 15/56$.

(2) Blue first, then Red: $(3/8) * (5/7) = 15/56$.

Total probability = $15/56 + 15/56 = 30/56 = 15/28$.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): If $P(A) = 0.4$ and $P(B) = 0.5$, and A and B are independent, find $P(A \cap B)$.

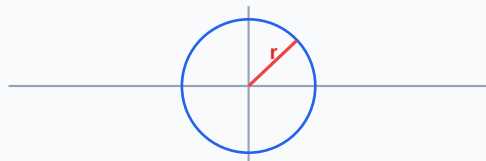
Q2 (2-Mark): Define mutually exclusive events.

Q3 (2-Mark): Write down the sample space when three fair coins are tossed.

Q4 (5-Mark): State total probability theorem.

Q5 (5-Mark): Explain Bayes' Theorem formula and its components.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Biostatistics

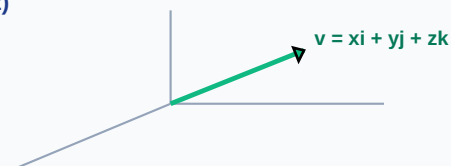
Bayes' theorem calculates diagnostic accuracy rates of clinical tests given false positive ratios.

Application Case: Asset Pricing

Probability distributions map investment volatility margins to construct diversified investment funds.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

